

ICT 2212 Ethical Hacking Labs (Weeks 1 – 13)

1. Objectives

- To work collaboratively and efficiently in your project team
- To complete the design and set-up of a reasonably secure server
- To be able to demonstrate successful foot-printing, scanning and enumeration of, and gaining of access to secure servers set up by other project teams

2. Team Information (per Lab Group)

Each team has 6 members, except 2 groups with 7 members.
The team list for all lab groups can be found on SIT xSiTe.

3. Attacking Scenario Objective

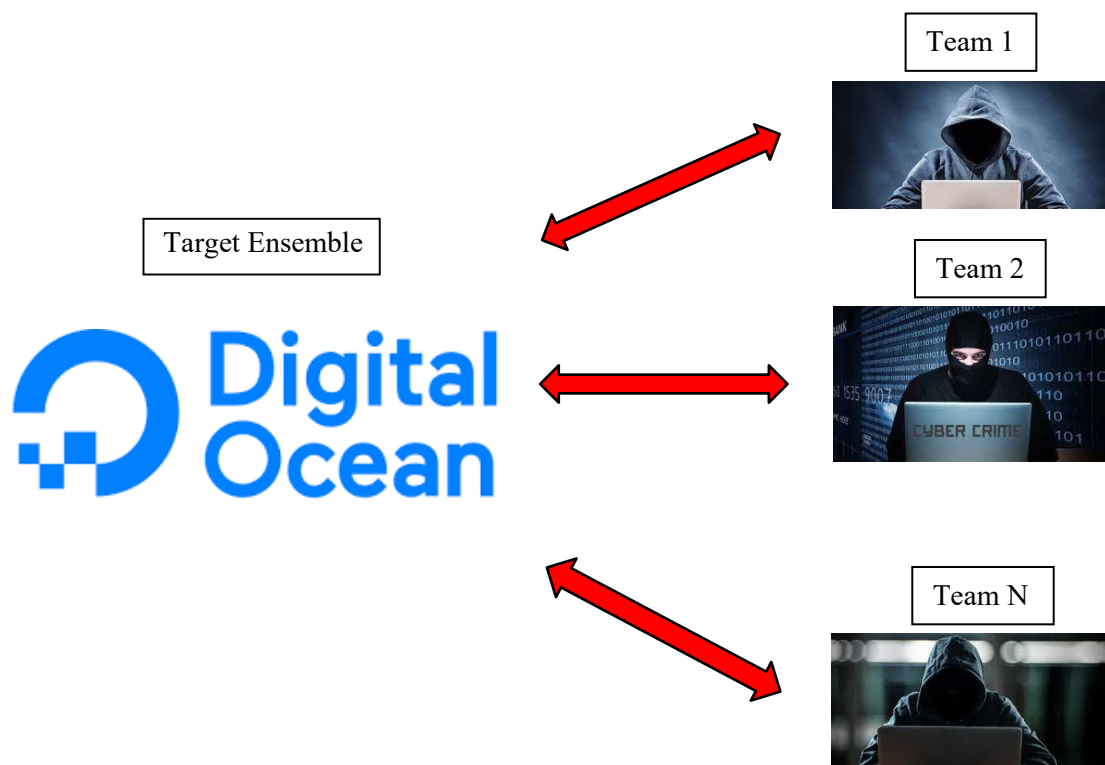


Figure 1

- The target ensemble contains all the servers set up by every project team in the lab group.
- Each server is assigned an IP address known only to the team who set it up.
- To score marks, each team must gather as much information via reconnaissance, scanning and enumeration and gain access to as many servers as possible in the target ensemble except their own server.

4. Coursework Deliverables Schedule

Week	1	2	3	4	5	6	7	8	9	10	11	12	13	14	CA deliverables
Week 1 - 6 Defender	Setup the own server														
				Harden and Demo the own server											
							R1								R1: Server Report due Week 7
Week 8 - 13 Attacker								Foot-print, Scan and Enumerate other servers (target) and Demo			Gain access to other servers (target) and Demo				Demo foot-printing, scanning and enumeration of target(s) Demo access gained to target(s)
												R2			R2: Penetration Testing Report due Week 12
													PR		PA: Peer assessment due Week 13

5. Coursework Assessment (CA) Deliverable Requirements

To be completed by every project team who has dual roles:

As a Defender:

- A team must design and set up a vulnerable server where the resource will be provided, at the latest by the end of Week 3 and hardening it by the end of Week 6. Each team's server will be assigned a unique IP address.
- The server system must be accessible via the Internet.
- The team must be able to demonstrate to the instructors that they can conduct successful foot-printing, scanning, enumeration of and gaining of access to their own server (NOT target) by the end of Week 6.
- After this demonstration, the defending team will freeze the system configuration and **keep running** the server system. Please don't stop or reboot the server system (i.e., the EC2 instance in AWS). Otherwise, the server system will be assigned a new IP address if start the server system again.
- The defending team's server IP address will serve as a target to other attacking teams.
- The defending team must **submit report ICT2212-R1 with the zip file name format of ICT2212-R1-Team XX (XX is the team number, report in word format, and software source codes)**, comprising the following, latest **by the end of Week 7 - 18 October 2026, 11:59pm**:
 - Team Number and Member Names
 - Steps of setup
 - Own server details e.g., IP address, server type, OS platform, vulnerable app, URL etc.
 - Hardening technique(s), tool(s) and process details
 - Vulnerability exploitation technique(s), tool(s) and process details
 - Software source codes
- The submitted results will be verified by the instructors (**If the submitted information is different from the server system, see Penalty section below**).

As an Attacker:

- Each team (as attacker) can attempt foot-printing, scanning and enumeration on **1 randomly assigned target** except your own server.
- All attacking teams shall complete their foot-printing, scanning and enumeration of their assigned targets by Week 10.

- All attacking teams must complete gaining of access to their assigned targets and demonstrate all to the instructors by Week 13, latest.
- The attacking team must **submit report ICT2212-R2 with the file name format of ICT2212-R2-Team XX (XX is the team number, word format)**, comprising the following, latest **by the end of Week 12 -22 Nov 2026, 11:59pm**:
 - Team Number and Member Names
 - Gaining Access – description of technique(s), tool(s) and process details
 - Maintaining Access – description of technique(s), tool(s) and process details
 - Covering Tracks – description of technique(s), tool(s) and process details

Scoring Information:

Each team will be assessed on how effectively it has performed in both defender and attacker roles. Marks will vary depending on successful exploitation(s) as well as the approaches of other successful attacker teams on the same targets and are allocated as follows:

- Setup, Hardening and Foot-printing, Scanning and Enumeration, and gain access to own server (max 5%)
- Own vulnerable server set-up report and software source codes, ICT2212-R1 (max 5%)
- Foot-printing, Scanning and Enumeration, and Gaining of access to assigned target(s) (max 7.5%)
- Penetration Testing Report, ICT2212-R2 (max 7.5%)
- Peer assessments are to gauge individual team member's contributions and teamwork

Penalty:

Any team found to have reported incorrect server information, deliberately “blocked” their system completely from foot-printing, scanning or enumeration and/or changed their system in any way without permission after week 6 will receive 0 marks for their R1 scores. The team will also have to restore their system to its original reported state.